



EE 530: Applied Optimization

Course Code	EE 530
Course Name	Applied Optimization
L-T-P-C	3-0-0-3
Prerequisites	Linear algebra (MA512 or IC 111) or Matrix Theory (EE522).
Intended for	UG /MS/MTech (CSP)/PhD/M.Sc. (Applied Mathematics)
Distribution	Core for MTech (CSP), Elective for B.Tech. III/IV year, MS, Ph.D, M.Sc (Applied Mathematics).
Semester	Even
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)
Revised	33rd BoA (Jan 2020) - renamed from “Optimization Theory”; 24th Senate (Feb 2020) - syllabus revised

Course Contents

- **Convex analysis:** convex sets, convex cones, polyhedral sets, extreme points and directions. Convex functions, properties and tests for convexity, operations that preserve convexity, conjugate function. [6 Lectures]
- **Convex optimization problems:** standard form, equality and inequality constraints, slack variables, eliminating equality and inequality constraints. Local and global optima. Optimality criterion for unconstrained, equality constrained and inequality constrained problems. [7 Lectures]
- **Module III:** Linear optimization problems with examples, linear and generalized linear-fractional programming. Quadratic problems with examples. Second order cone programming robust linear programming, linear programming with random constraints. Geometric programming with examples. Generalized inequality constraints conics form problems, semidefinite programming, examples. [14 Lectures]
- **Handling non-convexity:** Lagrangian duality theory Lagrangian dual function, strong and weak duality, duality gap. Certificate of suboptimality and stopping criteria, complementary slackness. KKT optimality conditions. Solving the primal via dual. [9 Lectures]
- **Applications of convex programming in communication and signal processing:** The choice of applications is left to the faculty member handling the course. Examples that could be used are: Optimal decentralized estimation (single and multisensor case). Pulse shaping filter design. Quasi-ML detection via SDP relaxation (or any other problems in CSP). [6 Lectures]

Text Books

1. S. Boyd and L. Vandenberghe, **Convex optimization**, Cambridge University Press, 2008.

References

1. M.S. Bazaraa, H.D. Sherali and C.M. Shetty, **Nonlinear Programming**, 3rd Edition, Wiley, 2006.
2. D. P. Bertsekas, **Nonlinear programming**, Athena Scientific, 1999.
3. D. G. Luenberger and Y. Ye, **Linear and nonlinear programming**, 3rd Edition, Springer, 2008.
4. Relevant literature.